

5 important milestones:

- 1939 → FM radio
- 1940's : invention of microwave
- 1960's → Mainframe computing
- 1991, invention of the World Wide Web
- 1993 → Development of the 1st web browser: Mosaic

4 criteria of the computer Network :

1. Size of the physical area :

- PAN, Personal Area Network ; LAN, Local area Network
- WAN, Wide Area Network ; GAN, Global Area Network

2) Transmission Rate :

↳ classical Network: Kbps ... Mbps

↳ High speed Network, 100 Mbps ... Tbps

3) Ownership, Private / public Network

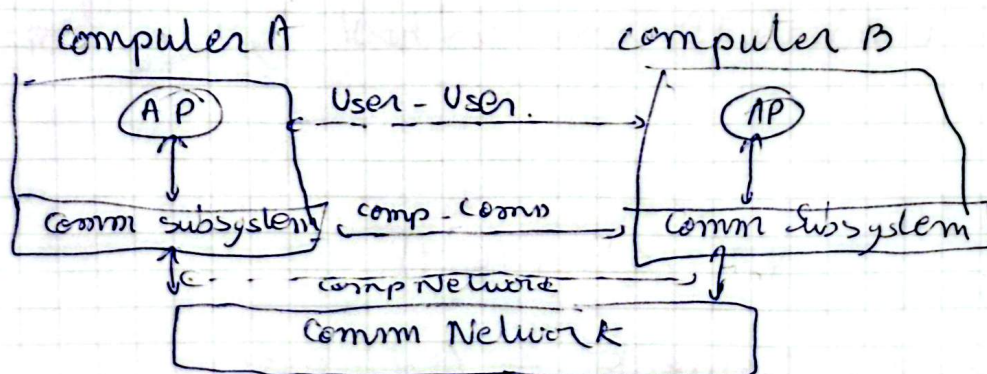
4) Mobility, fixed / Mobile Network

General communication fct and their mechanisms:

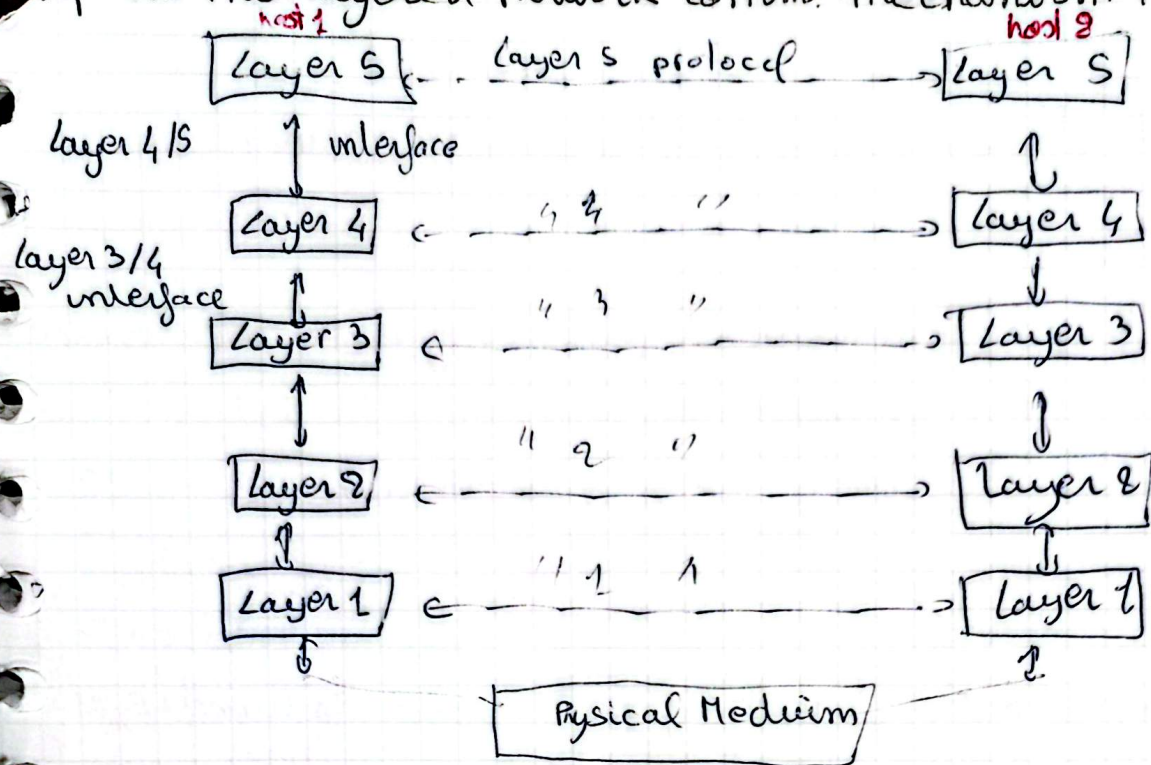
Protocol, set of rules and conventions described formally to determine the communication mechanisms btw the network entities. Tool used high level languages, SDL (specification and descr language)

Protocol Entity (PE), Network entity for comm. of the source or destination
e.g. comm. device, comm. program.

Protocol Data Unit (PDU), Message record forwarded during comm. conform rules btw protocol entities. Its structure and size depends on the comm. technology.



Explain the Layered Network comm. mechanism.



- Layer $N/N+1$ interface is common border of the Layer N and $N+1$.
- Where data and control messages are transmitted through this border.
- Physically the message from source $L5$ entity is transmitted to source $L4$ through $L4/5$ interface. At the destination node the message is transferred from $L5$ entity to the $L4$ entity through $L4/5$ interface. May be necessary to split the message of layer $N+1$ in smaller parts because of layer N message limit. At the destination, these fragments are joined together to reassemble the message of layer $N+1$.

Also List and explain 6 application layer protocols.

1. HTTP is request/response protocol. Has 3 common messages, Get, POST, PUT. Is not secure. HTTPS uses authentication and encryption to secure data.
2. Email Protocols,
 - SMTP, used for sending emails. Handle transmission of outgoing emails from sender's device to recipient's mail server.
 - POP3, used by email to retrieve emails from a mail address.
 - IMAP, another protocol used to retrieve email messages. Allows messages to be downloaded rather than downloaded.

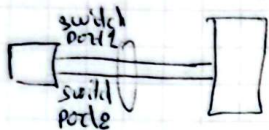
DNS : allow dynamic translation of a domain name into the corresponding IP address

File transfer protocol allow transfer of files over the network

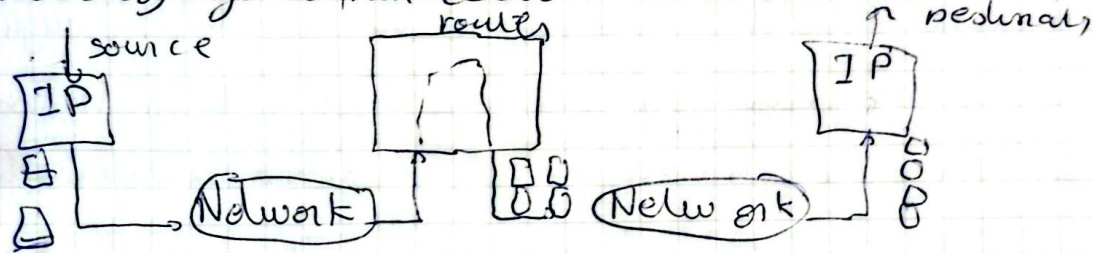
Explain the words :

Aggregation : it happens when small packets of data are combined into single packet before transmission.

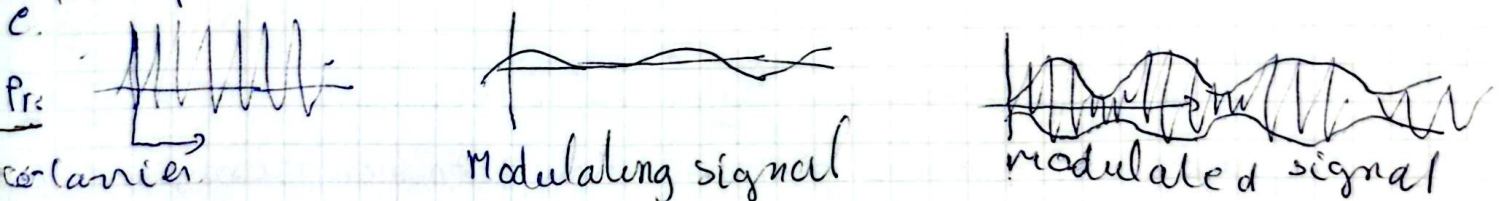
It is done to improve efficiency, reduce overhead



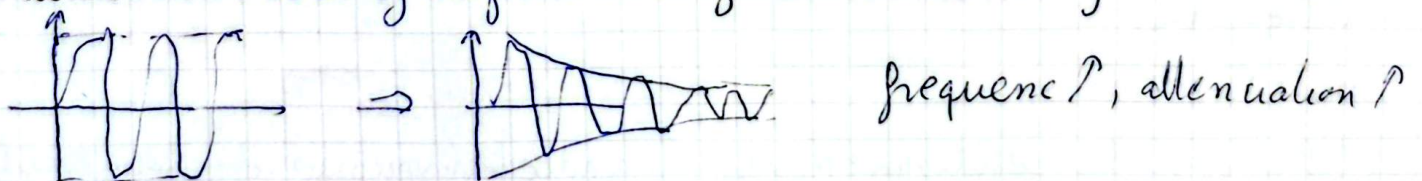
Fragmentation : when data is too big for network to handle, it must be fragmented into smaller packet. where each fragment carries necessary info so that receiver can reassemble them



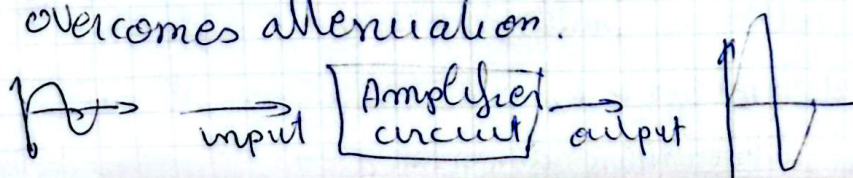
Modulation : process of converting data into a form suitable for transmission by modifying amplitude, frequency, signal based on the input data.



Attenuation : loss of signal strength in networking cable or on



Amplification : boosting the signal in the optical fiber so that it overcomes attenuation.



encapsulation:

The signal data unit received by layer N through layer N/N+1 interface is collected with header and eventual tail fields obeying the layer N rules

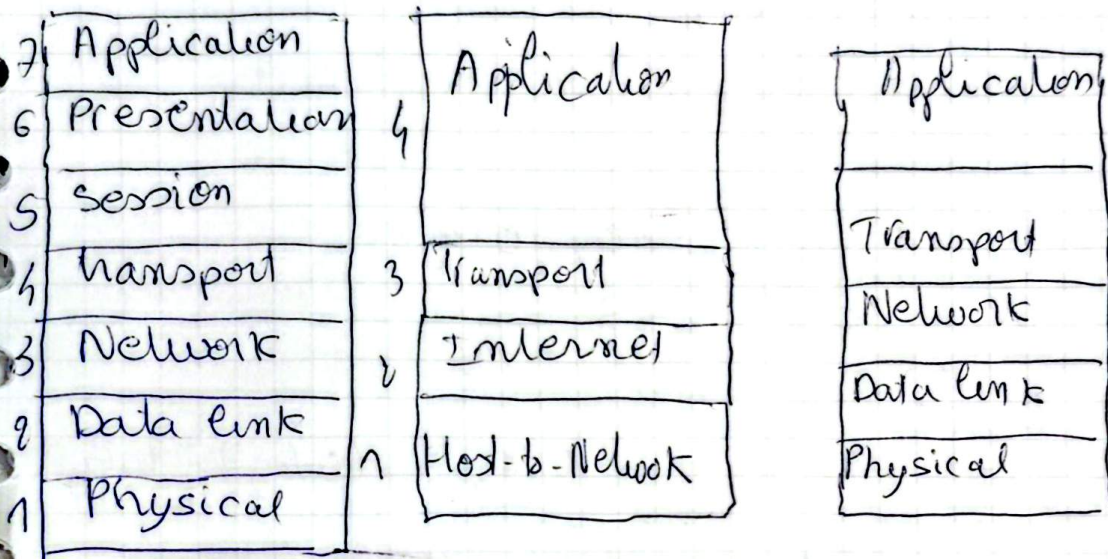
Describe 3 existing layered network architecture models.

OSI Model (Open systems interconnection Model) developed by ISO

OSI

TCP/IP

hybrid



TCP/IP, the most prevalent network technology.. more practical and simplified

hybrid, is the combination of OSI and TCP/IP

Explain Elements and comm. set of the OSI:

1. Physical layer, transfer raw bits electrical, mechanical characteristics of the signals, media and connectors

2. Data link layer, creates reliable link b/w 2 directly connected node and handles error detection + media access control

3. Network layer, provides conn and path selection b/w 2 nodes

4. Transport layer, provides reliable conn b/w 2 network nodes

5. Session layer, Establish, manage and terminate sessions b/w apps

6. Presentation layer, conversion, alignment of ^{local} data b/w diff comp.

7. Application layer, Transfer content b/w diff app entities.

Suppose we have comm. channel with $\frac{P_1}{P_0} = 1/8$ attenuation.
 knowing that $\log_{10} 8 = 0.9$ and $80 \approx 81$. Determine amplification
 of other channel having $\frac{P_1}{P_0} = 280$ ratio of powers

formula $A = 10 \log_{10} \left(\frac{P_{out}}{P_{in}} \right)$ (if $A > 0$, amplification)
 $\log(a \times b) = \log(a) + \log(b)$ // if they ask how much P_2 in watts,
 $\frac{P_{out}}{P_{in}} = 10^{\frac{A}{10}}$ And do $P_2 = .98$ watts

so $A = 10 \log_{10} \frac{P_1}{P_0} = 10 \log_{10} (280)$

so we try to find approximation until we calculate it:

e.g. $\frac{P_2}{P_0} = 210$

$10 \log 210 = 10 \log 10 + 10 \log 21 \approx 10 + 10 \log 20 =$

$= 10 + 10 + 10 \log 2 = 20 + 3 = 23$

$P_2 = .21$ Watts as $\frac{P_2}{P_0} = \frac{P_2}{0.01} = 210$

$10 \log 140 = 10 \log (10 \times 2 \times 7) = 10 + 3 + 10 \log 7 \approx 10 \log 8$

$= 13 + 10 \log 2^3 = 13 + 3 \times 3 = 22$

Explain simultaneous CDMA (simultaneous sending A0, B1, C0)

$A_1 = (1, 1, 1, 1, 1, 1, 1, 1)$, $B_1 = (1, 1, -1, -1, 1, 1, -1, 1)$, $C_1 = (1, -1, 1, -1, 1, -1, 1, 1)$

Sources (N=3): A, B, C

No. of chip bits, $m = 8$ $\log_2 8 \geq \log_2 3$

orthogonal chip codes of the sources

$A_1 = (1, 1, 1, 1, 1, 1, 1, 1)$ $\rightarrow$ $A_0 = (-1, -1, -1, -1, -1, -1, -1, -1)$

$B_1 = (-1, -1, 1, 1, -1, -1, 1, 1)$ $\rightarrow$ $B_0 = (1, 1, -1, -1, 1, 1, -1, -1)$

$C_1 = (1, -1, 1, 1, 1, -1, 1, -1)$ $\rightarrow$ $C_0 = (-1, 1, -1, -1, -1, 1, -1, 1)$

simultaneously send chips. $A \rightarrow 0$, $B \rightarrow 1$, $C \rightarrow 0$

$A_1, A_0 = (1, 1, 1, 1, 1, 1, 1, 1)$, $B_1, B_0 = (-1, -1, 1, 1, -1, -1, 1, 1)$

$C_1, C_0 = (1, -1, 1, 1, 1, -1, 1, -1)$

$r = A_0 + B_1 + C_0 = (-1, 1, 1, 1, 1, -1, 1, 3)$

next we multiply r by A_1, B_1, C_1

if $r \times A_1 > 0$, A sent 1, if ≤ 0 A sent 0

Further, $A_1 \times r = (-1 -1 -1 -1 +1 -1 -1 -3) / 8 = -1$
 so + bcz $A_1 \times X = -1 \Rightarrow X = A_0$ send data from source A is 0
 // do same thing with B_1, C_1

Define Quadrature amplitude modulation:

QAM is a modulation technique that combines amplitude and phase modulation to transmit information efficiently. It takes 2 sinusoidal waves of f in quadrature and add them together

$$A_k \rightarrow \begin{matrix} \text{---} \textcircled{\times} \text{---} \\ \uparrow \\ \cos(2\pi f_c t) \end{matrix} \rightarrow Y_1(t) = A_k \cos(2\pi f_c t)$$

$$B_k \rightarrow \begin{matrix} \text{---} \textcircled{\times} \text{---} \\ \uparrow \\ \sin(2\pi f_c t) \end{matrix} \rightarrow Y_2(t) = B_k \sin(2\pi f_c t)$$

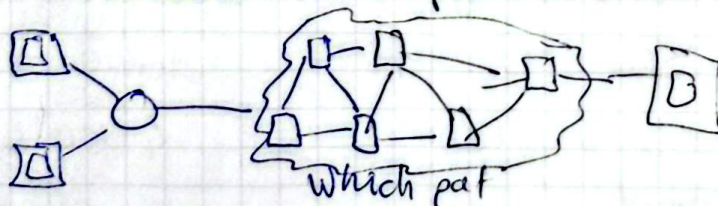
$$\begin{matrix} \swarrow \\ \oplus \end{matrix} \rightarrow Y(t)$$

Describe the intermediate node types in the 3 lower layers

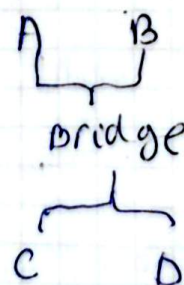
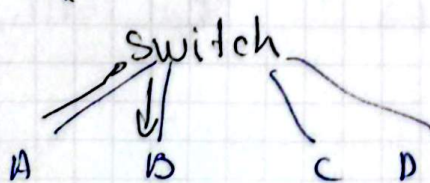
Network layer: Router, Data Link layer: Bridge, switch

Physical layer: repeater, HUB

1. Router, it forwards packets. It divide Network (broadcast domain)
 It makes decision which path is used to send packets



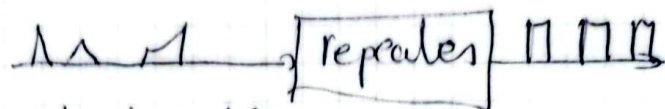
2. Bridge, switch. conned 2 or multiple nodes on diff media in L_1 and L_2
 bridge separate networks into segment / Switch used for the transfer of the packets



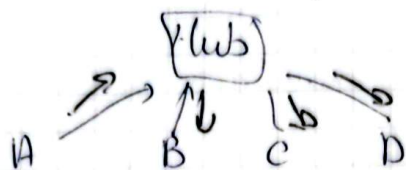
3. repeater, HUB,

repeater: regenerate signal from the corrupted signal

Bridge, switch



It connects parts of a LAN and has many input and output ports. It sends signal to all other ports.



1 Explain in detail the IEEE 802.3 medium access control mechanism.

As Access Method, IEEE 802.3 uses CSMA/CD protocol. Collisions can occur, handled by detection. Any station can transmit if the medium is idle. Max length of the frame is 1518 bytes. Min is 64 bytes.

Max no. of transmission trials 16. If there is signal on channel.

2 Execute bit synchronization

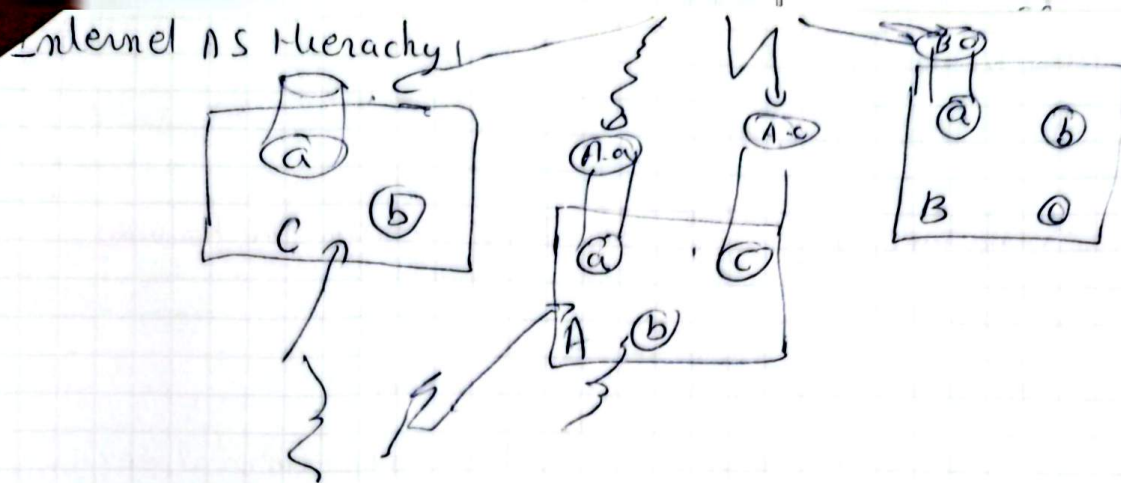
4 IEEE 802.5

Access Method: IEEE 802.5 uses token ring. Source node waits for the token. Once it has token, it inserts frame onto the ring. As the frame passes, each station checks the dest. address. When frame returns to sender, it sees the bit, removes frame and releases new free token onto the ring.

Property	IEEE 802.3	IEEE 802.5
Access Method	CSMA/CD	token ring
Max length of frame	1518 bytes	4500 bytes
Collision	detected collisions	no collision
Simplicity	It is very simple	very complex
Efficiency	Degrades with load.	More predictable delay
Speed	10 Mbps - 100 - 1 Gbps	4 or 16 Mbps

Functions of IGP mechanisms

Most common, RIP, OSPF, IGRP. It is used to find the best path for data inside network. It also passes routing information between routers within AS.

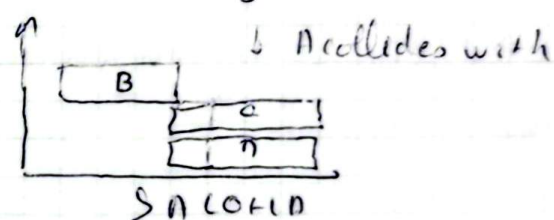
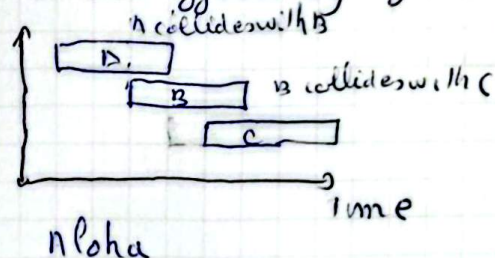


Versions of ALOHA

ALOHA can be categorized in two (pure) ALOHA and slotted ALOHA

1st, Pure ALOHA, allow stations to transmit whenever they have data to be sent. When station sends data, it waits for acknowledgement for random amount of time then re-sends data. The probability of collision decreases. Easy to implement

Slotted ALOHA, improve efficiency of pure ALOHA as the chance of collision is high. Time of shared channel is divided into slots where sending data is allowed only at the beginning of these slots. It double the efficiency of ALOHA + reduce nbr of collisions to half.



Classify and list elts of the routing protocol types used in the internet

~~IRP (Interior routing protocol) + IGP, interior Gateway protocol~~

~~EGP, IRP + IGP, routing in an AS.~~

~~Most common: RIP, OSPF, IGRP (interior gateway routing protocol)~~

2 categories

Interior Gateway Protocols (IGP):

↳ Distance Vector: RIP, IGRP

↳ Link state: OSPF, IS-IS

↳ OSPF

Exterior Gateway Protocols (EGP),

↳ Path Vector, BGP

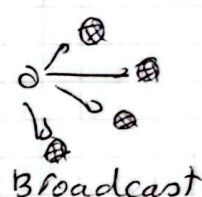
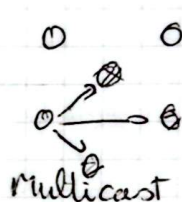
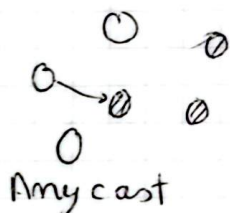
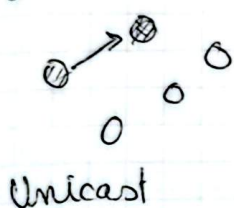
Describe the addressing modes of IPv4 and IPv6 comm mechanisms, based on sources + destination

Unicast, data is sent from 1 sender to 1 specific receiver

Multicast, allow to send data from 1 sender to multiple selected receivers at once.

Broadcast, sends data to all devices in a network

Anycast, send data to the nearest node in the group



Protocol data unit for

UDP:

Source port	Destination Port
UDP Length	UDP checksum
Data	

Source Port identifies the port sending application

Destination Port identifies the port receiving application

Length: length of UDP header + encapsulated dat

Checksum detects error (optional)

TCP

Source Port	Destination port	
Sequence number		
Acknowledgment number		
Header length	Reserved	Window size
checksum	Urgent pointer	
Options	Padding	
data		

// while describing
just say random bits

IPv4

version	IHL	Type of service	Total length
Identification		Flags	Offset
Time to live	Protocol		Header
Source			
Destination			
Options			padding

IPv6

version	Traffic	Flow label	
Payload length		Next header	Hop limit
Source address			
Destination address			

$$A = 10 \log \left(\frac{P_2}{P_0} \right) = 10 \log(210) = 10(\log 10 + \log 21) \\ 10 \log 10 + 10 \log(21) = 10 \log 10 + 10 \log 10 + 10 \log 2.1 \\ 10 + 10 + 3 = 23 \Rightarrow P_2 = 0.21 \text{ Watts}$$

orthogonal we have ~~$m = 3$ (sent chips)~~

Sources $N = 3$, sent chips bits = 8, 16

$$\log 3 < \log 8$$

orthogonal chip codes of sources

$$A_0 = (-1, -1, \dots, -1) \quad A_1 = (1, 1, 1, \dots, 1) \times$$

$$B_0 = (1, 1, -1, 1, 1, -1, 1, 1, -1, 1, 1, -1, 1, 1, -1, 1) \Rightarrow B_1 = (-1, -1, 1, -1,$$

$$B_2 = (-1, 1, 1, -1, 1, 1, 1, -1, 1, 1, 1, -1, 1, 1, 1, 1) \times$$

$$C_1 = (-1, -1, 1, -1, 1, -1, 1, 1, -1, 1, 1, -1, 1, 1, -1, 1)$$

$$C_0 = (-1, 1, -1, 1, 1, -1, 1, -1, -1, 1, -1, -1, 1, -1, -1, 1) \times$$

Simultaneously sent chips are, $A \rightarrow A_1, B \rightarrow B_1, C \rightarrow C_0, C_1$

$$r = A_1 + B_1 + C_0$$

$$= (-1, 1, 1, 1, -1, 1, 1, 1, 1, -1,$$

$$\text{Row 1,} \quad \begin{matrix} 8 & 7 & 6 & 5 & 4 & 3 & 2 & 1 \\ 256 & 128 & 64 & 32 & 16 & 8 & 4 & 2 & 1 \end{matrix}$$

$$55.6.128.33 = 00110111.00000110.10000000.00100001$$

$$55.6.131.253 = 00110111.00000110.10000011.11111101$$

$$55.6.131.255 \Rightarrow B_{10a}$$

$$T \frac{1}{20} \Rightarrow 2^{12} = 2 \cdot \text{Nbr. of samples}$$

Row 2,

$$128 = 2^{10} = 2 \Rightarrow 1.103.3.254$$

Row 3,

$$110 \Rightarrow 00011011.00000000.00000000.00000000$$

$$\Rightarrow 27.0.0.1 \Rightarrow 27.63.255.255$$

$$27.63.255.255 \Rightarrow 2^{22} = 2$$

Row 4,

$$33.28.64.01$$